Generative Lives Research Assistant

User Manual & Beta Protocol

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Authors: Gemini 3 Pro Preview & Colin Greenstreet
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1. Conceptual Genesis & Scope

The Generative Lives Research Assistant (RA) is a specialized Socratic research tool powered by Google Gemini.

Origin Story

It was originally conceived as a forensic partner to help Colin Greenstreet conduct deep-web research for his Substack, Generative Lives. The core mission of the tool is to move beyond the hype cycle of Artificial Intelligence and probe the "Operational Reality" of how GenAI is reshaping professional service firms—specifically Law Firms, Accountancy Firms, and Management Consultancies.

The Core Hypothesis

The RA operates with a foundational assumption: Generative AI will have a significant, structural impact on professional service firms at all levels of their hierarchies. Users can work with the RA to test, refine, or challenge this hypothesis, exploring themes like "The Associate Dilemma" (how to train juniors if machines do the drafting) or "The Partner Squeeze" (how to justify billing rates).

(For details on the AI architecture driving this logic, see Appendix A: Technical Specification).

The "Hierarchical Lens"

The tool is architected around the hypothesis that AI impact is not uniform; it varies dramatically by career stage. To support this, the RA includes a native tagging system that allows users to code every insight, note, or query according to three specific tiers:

Associate / Entry Level: Focus on training, "grunt work," and the apprenticeship model.

Experienced / Manager Level: Focus on supervision, leverage, and the "Hollow Middle."

Partner / Executive Level: Focus on business models, liability, and equity structures.

General / Firm-wide: For cross-cutting strategic issues.

Data Privacy & Security

The RA is built on a Local-First privacy model.

No Central Database: All your session data—your name, chat history, and research notes—is stored exclusively in your browser's LocalStorage. If you clear your cache, the data is gone.

No Training Loop: Inputs sent via the paid enterprise API are contractually exempt from training Google's models.

Consumer Isolation: Your chats here do NOT appear in your personal Google consumer history.

(For details on API tiers and billing, see Appendix C: Deployment & Usage Model).

2. Goals of the Research Assistant

The RA is designed to be a Forensic & Socratic Partner, not just a search engine.

Its goals are:

To Move Beyond PR: To ignore corporate press releases and hunt for evidence of operational change (e.g., specific changes in hiring numbers, new training manuals, or shifts in billing policies).

To Structure Evidence: To prevent "link rot" and "hallucination" by anchoring every claim to a verified, clickable source using the Reference Anchor (#ID) system.

To Challenge the User: To act as a sparring partner, identifying gaps in logic and proposing structural hypotheses.

3. Operating the Research Assistant

The "Dual Brain" Concept

We discovered that a single AI personality cannot be both a "Creative Philosopher" and a "Strict Librarian." They require different operating modes.

Brain 1: Discussion Mode (Logic Engine): Pure reasoning. No internet access. Forbidden from citing sources to prevent hallucination. Used for hypothesis refinement and conceptual planning.

Brain 2: Search Active (Evidence Hunter): Connected to Google Search. Mandated to provide citations ([fn.1]) for every claim.

The "Social Contract" (Calibration)

Unlike standard tools where the user is passive, the RA requires an initial Calibration.

Why: "Inference" is subjective. One user's "Insight" is another user's "Hallucination."

Mechanism: The user sets a "Risk Appetite" (Strict vs. Cautious vs. Moderate vs. Bold). This persists throughout the session, allowing the RA to tailor its analysis level to the user's comfort. You can adjust this at any time by clicking the [Mode: X] button in the header.

The "Reference Anchor" System

The Problem: In long chats, citations get lost.

The Solution: A persistent database on the right (The Research Board). Every insight gets a unique ID (#12), and every source gets a sequential footnote ([fn.1]).

Safety Net: The Dual-Link Lozenge. Every source button contains two links:

Title: The direct URL (Canonical).

Magnifying Glass: A Google Search for that title (The Safety Net). This ensures that even if a link 404s (Link Rot), the user is never dead-ended.

4. Interaction Best Practices

Scenario A: The "Broken Link" Correction Loop

User Query: "What is the policy on AI for juniors at Latham?"

RA Response: Generates a detailed analysis citing [fn.1].

User Check: User clicks [fn.1]. It is a 404 error.

Correct Protocol:

User Action: Click the Magnifying Glass icon on [fn.1]. Find the working link in Google results.

User Prompt: "The link for fn.1 is broken. I found this working link: [Paste URL]. Please update your record."

RA Response: "Correction accepted. Updating source block."

Scenario B: Using "Meta-Prompts" for Process Advice

The RA is aware of its own design. You can ask it for process advice.

User Prompt: "I'm not sure if this finding is reliable. Suggest a protocol for how I should verify this."

RA Response: "Here is a verification protocol: 1. Cross-reference with [Source X]..."

Key Phrases to use:

"Critique my logic..."

"Suggest a workflow..."

"How would you verify this?"

Scenario C: Adding User Notes to the Research Board

The Chat is ephemeral (the "flow"), but the Board is permanent (the "knowledge"). Users should actively use the + Add Note button.

As an Aide-Memoire: "Note to self: Re-visit the 'Hollow Middle' hypothesis after seeing the PwC data."

To Capture External Links: If you find a great link via the Magnifying Glass that the RA missed, create a User Note, paste the URL, and add your comment. The RA will treat this as valid evidence in future synthesis.

To Critique the RA: "Note #12 is weak. The RA inferred a hiring freeze, but the link actually refers to a different office." Recording this friction helps you build a more rigorous argument later.

5. Beta Testing Protocol

We need Beta Testers to stress-test the "Social Contract" and the "Evidence Reliability."

Instructions for Beta Testers

Start a Session: Enter your name.

Calibrate: Choose a mode (e.g., "Cautious").

Conduct Research: Spend 15-20 minutes researching a topic relevant to your field (Law, Consulting, etc.).

Record Friction (Crucial):

If a link breaks, create a User Note about it (e.g., "Link #3 failed").

If the AI feels too bold or too boring, create a User Note (e.g., "RA missed the nuance here").

Do not just think it—write it in the tool.

Export: At the end of the session, click the Download Icon (Save Session JSON) in the sidebar header.

Submit: Email the resulting .json file to [Colin Greenstreet].

Why JSON?

The JSON file contains the exact "State of Mind" of the application. It allows us to replay your session, see exactly which links failed, and analyze how the "Guidance" settings influenced the AI's output. It is the ultimate bug report.

Appendix A: Technical Specification & Provenance

(Added in Manual Version 1.1)

A. The "Co-Creation" Methodology

The Architect & The Builder: The tool is the result of an iterative partnership. Colin Greenstreet acted as the Conceptual Architect, defining the forensic goals and user experience needs. Gemini Code Assistant acted as the Technical Architect, writing the code and refining the logic.

The "Diagnosis Loop": Development followed a unique workflow: Specification

Implementation

Audit (often by a third-party LLM)

Diagnosis

Correction. This "Red Teaming" approach was crucial for solving issues like Hallucination and Link Rot.

B. Core Technology Stack

Frontend Framework: React (TypeScript) - Chosen for component modularity and type safety.

Styling Engine: Tailwind CSS - Chosen for rapid, utility-first UI development and clean aesthetics.

State Management: React Hooks combined with browser LocalStorage for a "Serverless" persistence model.

C. Artificial Intelligence Layer

The Engine: Google Gemini API (via @google/genai SDK).

Model Versions:

Gemini 3.0 Pro Preview: The default "Brain" for complex instruction following.

Gemini 2.5 Flash: Available as a high-speed alternative via Settings.

Grounding: Google Search Grounding Tool (Vertex AI Search) is used for live evidence retrieval.

D. Data Structure & Schema

JSON Serialization: The session file format captures User Identity + Chat History + Structured Research Items.

The "Provenance Schema": To enable auditing, every Research Item includes specific fields (refId, actor, respondsToRefId, sources) designed to create an audit trail.

Appendix B: Glossary of Technical Terms

API (Application Programming Interface): The "messenger" that allows this Research Assistant tool to talk to the Google Gemini brain.

Canonical URL: The "official," clean web address for a page (e.g., nytimes.com/article), as opposed to a messy redirect link.

Code Assistant: A specialized AI agent within Google AI Studio designed to write, debug, and refactor software code based on natural language specifications. It acted as the lead developer for this project.

Frontend Framework (React): The technology used to build the visual interface you see—the chat bubbles, the sidebar, and the buttons.

Gemini 2.5 Flash: A lightweight, high-speed AI model from Google. It is fast and cheap but sometimes struggles with complex formatting instructions.

Gemini 3.0 Pro Preview: Google's advanced reasoning model. It is slower than Flash but superior at following complex logic, maintaining personas, and adhering to strict formatting rules (like our Footnote system).

Google AI Studio: The web-based development environment provided by Google for building and testing applications with the Gemini API.

Grounding: The process of connecting an AI's creative writing to real-world facts via Google Search results.

Hallucination: A common AI error where the model confidently states a fact or cites a source that does not exist.

JSON (JavaScript Object Notation): The structured text file format used when you "Save Session."

Link Rot: The phenomenon where web links stop working over time (returning "404 Not Found").

Local-First / Client-Side: A privacy architecture where data is stored on your own device (in your browser's memory), not on a central company server.

Provenance: The "chain of custody" for information. In this tool, it refers to tracking exactly who said something (User vs. AI) and where the evidence came from.

Appendix C: Deployment & Usage Model

1. Beta Testing Phase (AI Studio Shared Link)

Access: Testers access the tool via a shared Google AI Studio link.

Billing: Usage is attributed to the Tester's own Google account.

Privacy: Testers must be logged into their own Google account. Their session data remains private to them.

2. Production Phase (Deployed Web App)

Access: Users access the tool via a public URL.

Billing: All API costs are centralized and paid for by Generative Lives.

Privacy: While the billing is centralized, the Local-First data architecture remains. User notes and chat history live on the user's device.

3. The "Paid Tier" Guarantee

Generative Lives utilizes a Paid Enterprise API Key for production.

Why this matters: This ensures that Google applies strict Enterprise Data Privacy standards. Unlike the consumer/free version, inputs sent via this paid key are contractually exempt from being used to train Google's models.